

$$\mathcal{L}_e = \mathcal{L}_{\text{MSE}} + w(e) \mathcal{L}_{\text{inv}} + 0.1 \mathcal{L}_{\text{lat}}, \quad w(e) = \min\left(1, \frac{e+1}{10}\right)$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{BF} \sum_{b=1}^B \sum_{f=1}^F (\hat{y}_{b,f} - y_{b,f})^2, \quad \hat{y} = \Psi \mathbf{b}(\mathbf{a}) + \psi_0$$

$$\mathcal{L}_{\text{inv}} = \text{Huber}(\hat{\mathbf{a}}, \mathbf{a}), \quad (\hat{\mathbf{a}}, \hat{\mathbf{z}}) = T^{-1}(\mathbf{b}^*), \quad \mathbf{b}^* = \frac{1}{F} \Psi^\top (y - \psi_0)$$

$$\mathcal{L}_{\text{lat}} = \text{Huber}(\hat{\mathbf{z}}, \mathbf{0}) \text{ (padded variants; absent for Q8, where } Q = D)$$